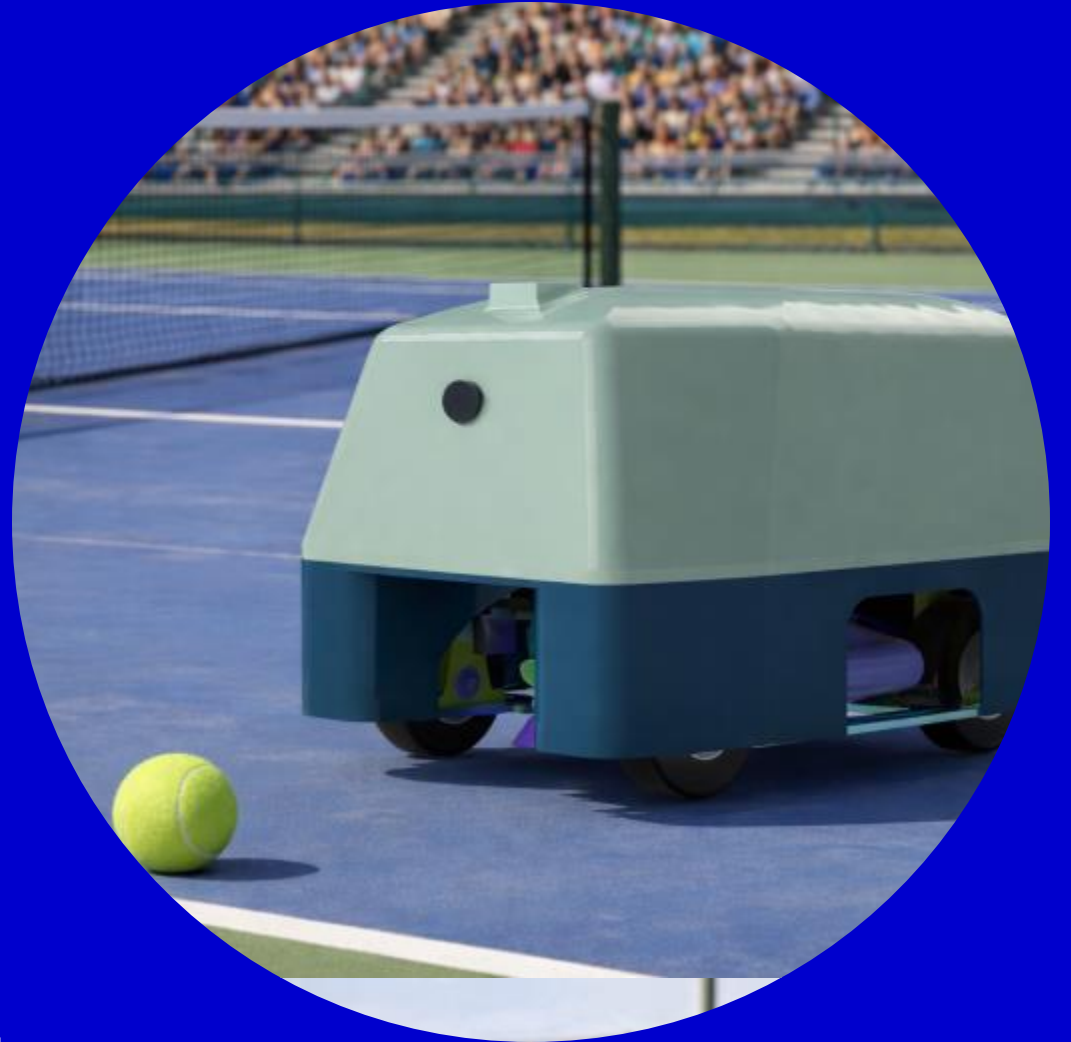


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SHOBOT, A Tennis Ball Collecting Robot

By Team ECHO

Members: Eito Ogawa, Charles Hung, Hengze Li, Oscar Chan



PROJECT AIM

To develop a tennis ball collecting robot to replace ball kids in a competitive match

CUSTOMER SPECIFICATION

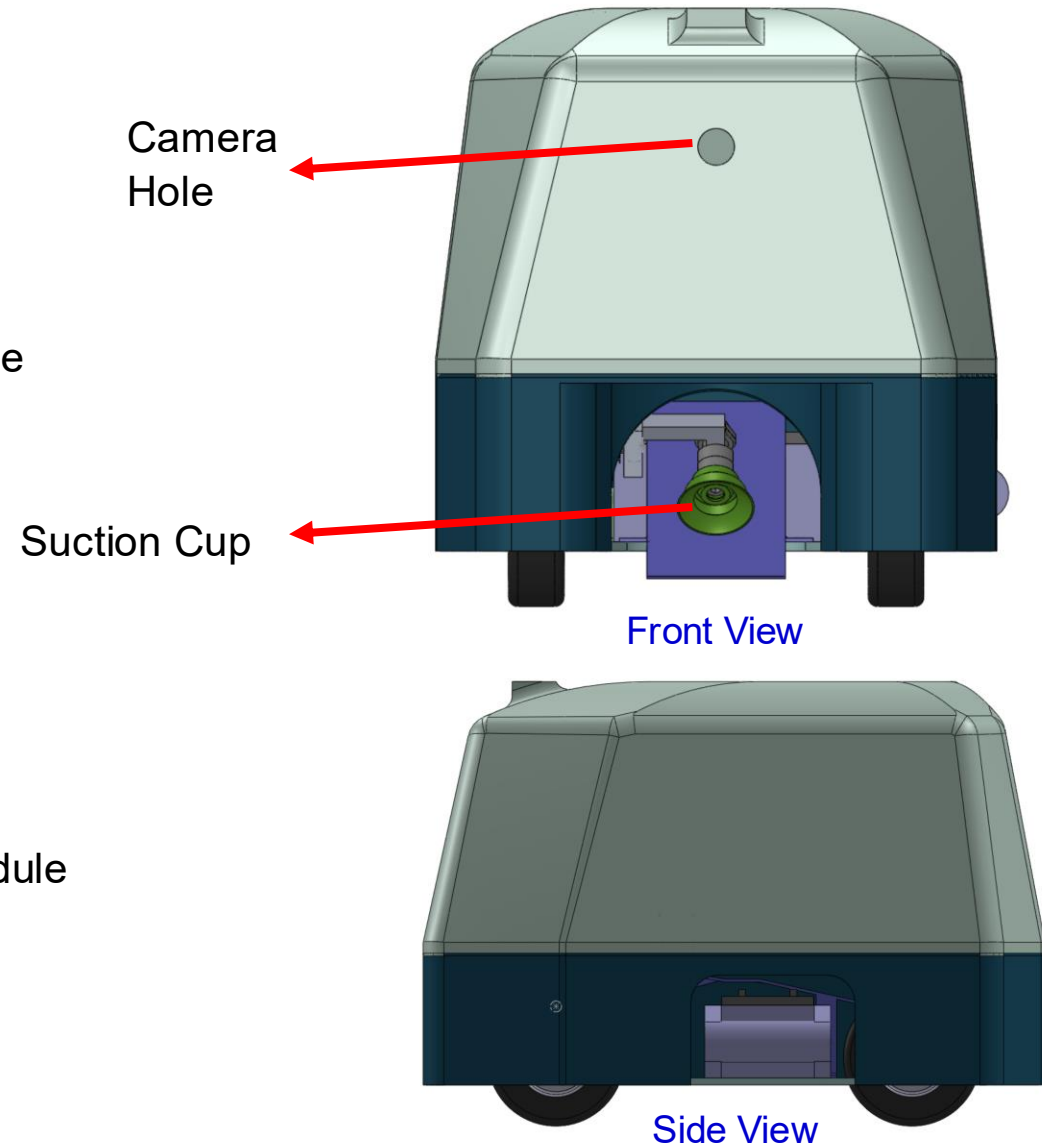
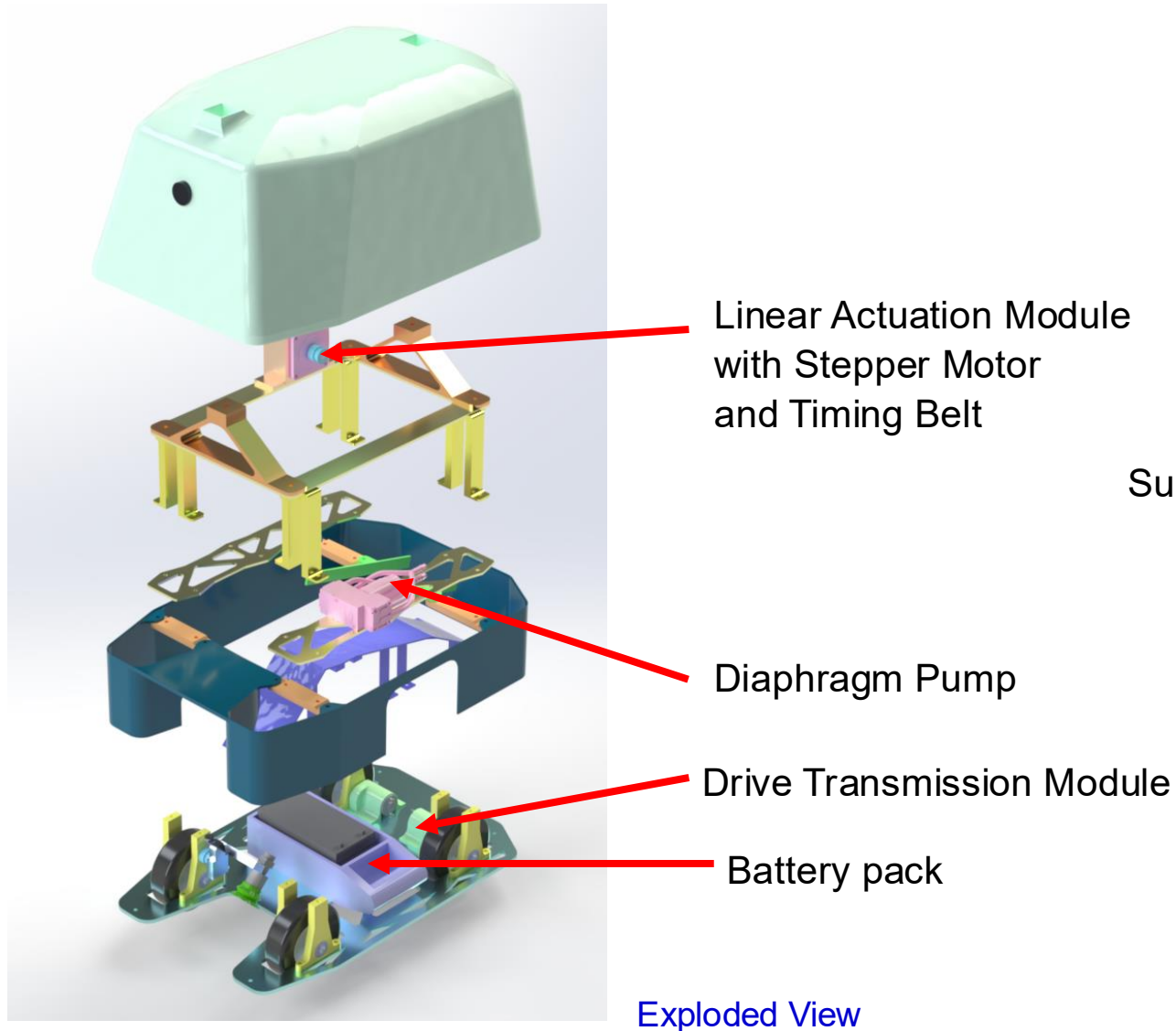
- ☐ To use a **vacuuming mechanism**
- ☐ On and off court **quickly**
- ☐ **Aesthetically pleasing** design
- ☐ Production run of **10000 products**
- ☐ Use a rechargeable battery
- ☐ Sustainability-conscious product

PRODUCT DESIGN SPECIFICATION

PRODUCT DESIGN SPECIFICATION		
Element		Criteria
Customer	Aesthetics	Simple casing geometry; Minimalist to not be too distracting for players.
Operation	Performance	Max 10 seconds for whole cycle
	Environment	Can be utilised indoors and outdoors
	Size	300X450X300mm
	Weight	7kg
	Sustainability	60Kg CO2 per year
Life	Product Life	10 years
	Service Life	6h
	Maintenance	Corresponding faulty components can be swapped out and replaced
Producer	Quantity	From client request
Regulatory	Safety	Minimise risk of collision and battery malfunctions
	End of Life	Modular design eases disassembly so that it can be recycled

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Our Design Solution: SHOBOT

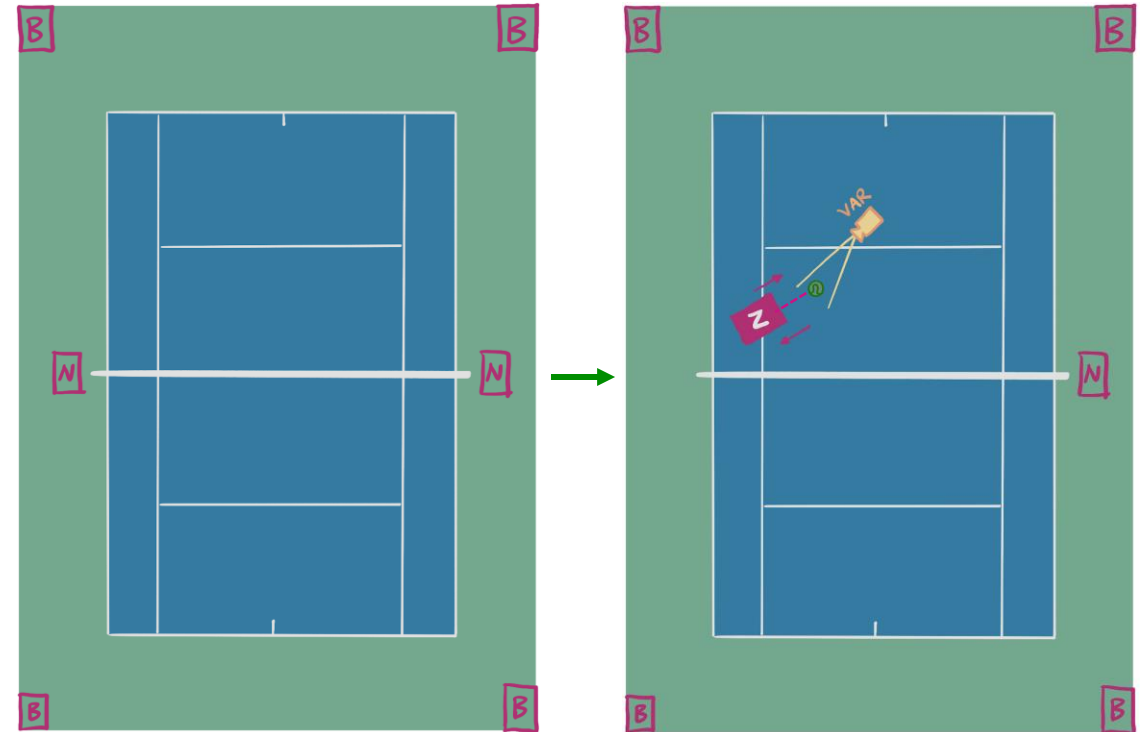


Ball Retrieval Mechanism



1. Nearest robot detects and moves to a ball
2. Ball enters mouth of robot
3. Suction cup lowers down the ramp
4. Suction cup turns on – ball is gripped
[Aided by **latch**]
5. Ball is pulled up ramp
6. Vacuum turned off → Ball released in storage unit
7. Robot moves back to docking station

User



Tennis Court

- Fully autonomous → Sensors and control by Electronics Team
- One robot in each corner (*Base*), one by each side net (*Net*)
- Global ball tracking by VAR; local by LiDAR/camera sensors

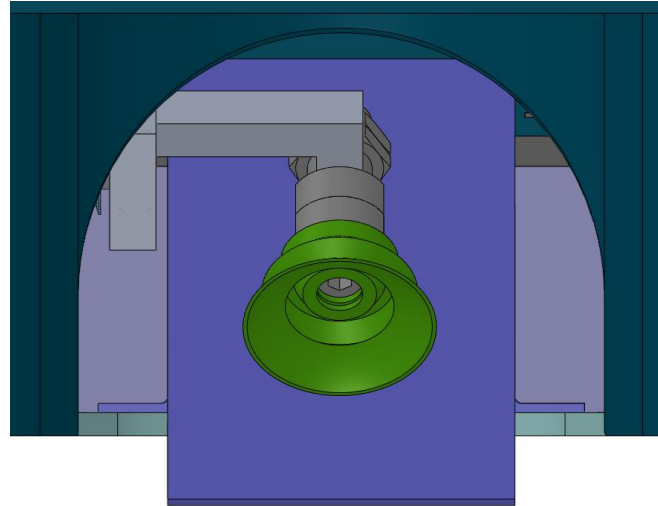
Suction Gripping Over Suction Swallowing

Require a lot of power

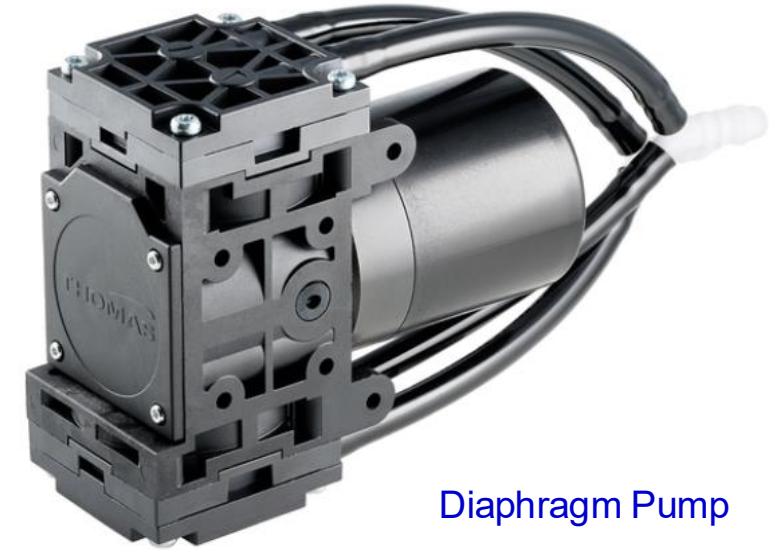
Large motor size

Debris got suck in

Loud vacuum noise



Suction Cup and Ramp



Diaphragm Pump

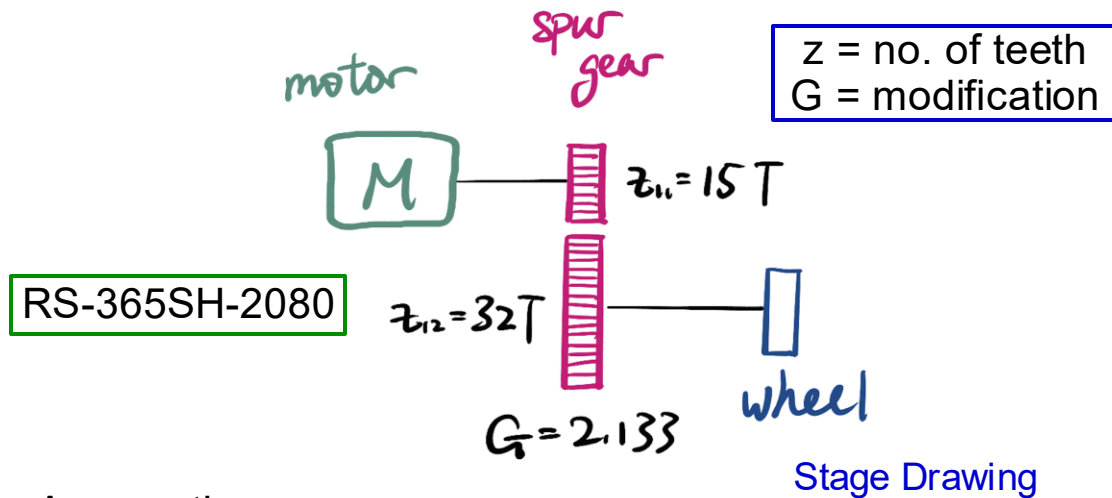
Diaphragm Pump Over Fan Vacuum

- ❑ Diaphragm pump can maintain strong negative vacuum at low airflows; A fan-based pump is highly sensitive to sudden airflow changes, leading to unstable suction.
- ❑ They are self limiting → less power and will not overheat
- ❑ Small size ; low weight ; less noise
- ❑ Can handle air-dust mixtures

Slanted Ramp Over Vertical Ramp

- ❑ Reduces electronics complexity
- ❑ Less force required to grip and move tennis ball

Wheel Drivetrain



Assumptions:

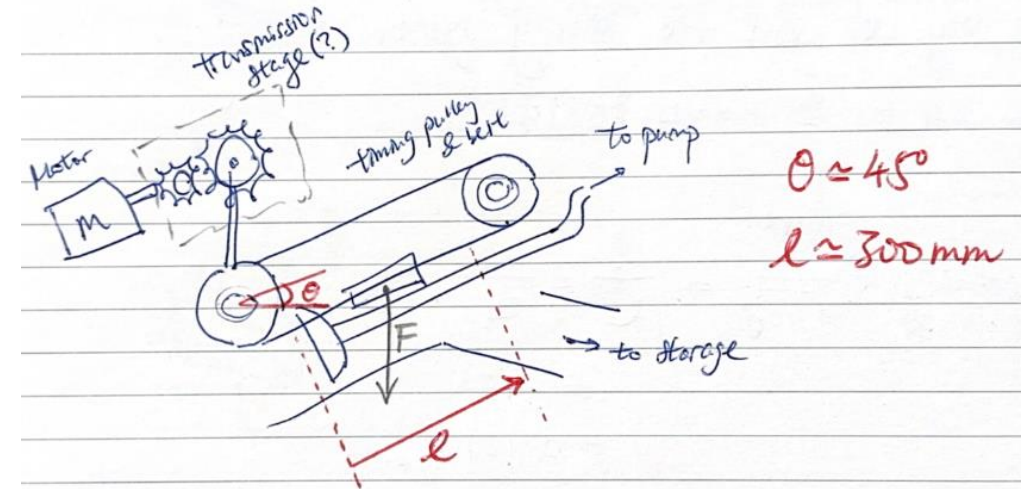
- [Constraint] 10s round trip – target velocity 9.92 km/h
- [Constraint] Resistive torque of 61 mNm
- Acceleration time of 0.4s

No load speed of 16400 rpm → Achieved 9.69 km/h

Module 0.5 Steel EN8 Gears → Small but durable

Pinion Safety Factor: 8.5

Linear Actuation module



Assumptions:

- Slope angle θ , Travel distance l
- Mass $m = 150\text{g}$, Travel time $t_{\max} = 2\text{s}$

$G = 230.44...$ impossible!

→ Stepper motor

→ HTB rated belt power 94W, $SF_{\text{power}} = 9.8$

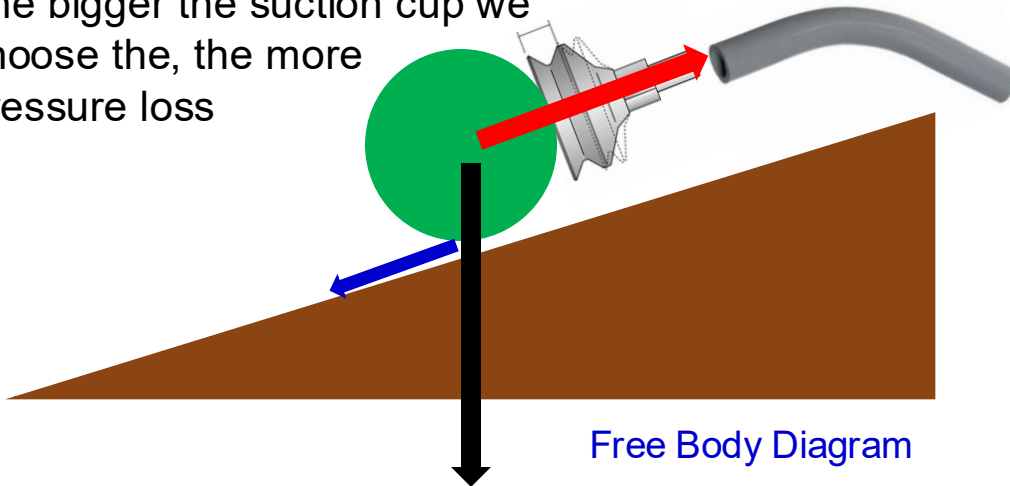
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Pump selection

1. Assume airflow (Conservative) when suction cup touches the ball
2. Find out maximum available negative pressure generated by pump
3. Estimate the pressure drop due to bending, suction cup
4. Calculate if force is enough to overcome gravity and friction.

$$F_{suction} = \Delta P * A_{eff\ area}$$

The bigger the suction cup we choose the, the more pressure loss



Suction cup selection

The inlet diameter smaller than a tennis ball rim seal to conform to rough surfaces to minimize leakage.

Technical Analysis: Miscellaneous

Material Selection

ASA

Acrylonitrile Styrene
Acrylate

Outer frame + Ramp
UV and weather resistant for outer frame
lightweight

Aluminum

A356-T6

Chassis + Structural Frame:
high strength-to-weight ratio
corrosion resistance, recyclability

Low-Alloy Steel

AISI 4140

Shaft
high strength allows a smaller shaft
diameter, fitting the tight packaging space

Selected using Ansys Granta Edupack software.

Requires:

BATTERY SELECTION

- ☐ 12 V
- ☐ Minimum 10 A discharge current for motors selected
- ☐ Uses ~0.8 Ah* for every 1000 ball captures (typical tennis game).

*Including additional electronics component

Selected 12V, 3.2Ah Lead Rechargeable Battery
From RS Components

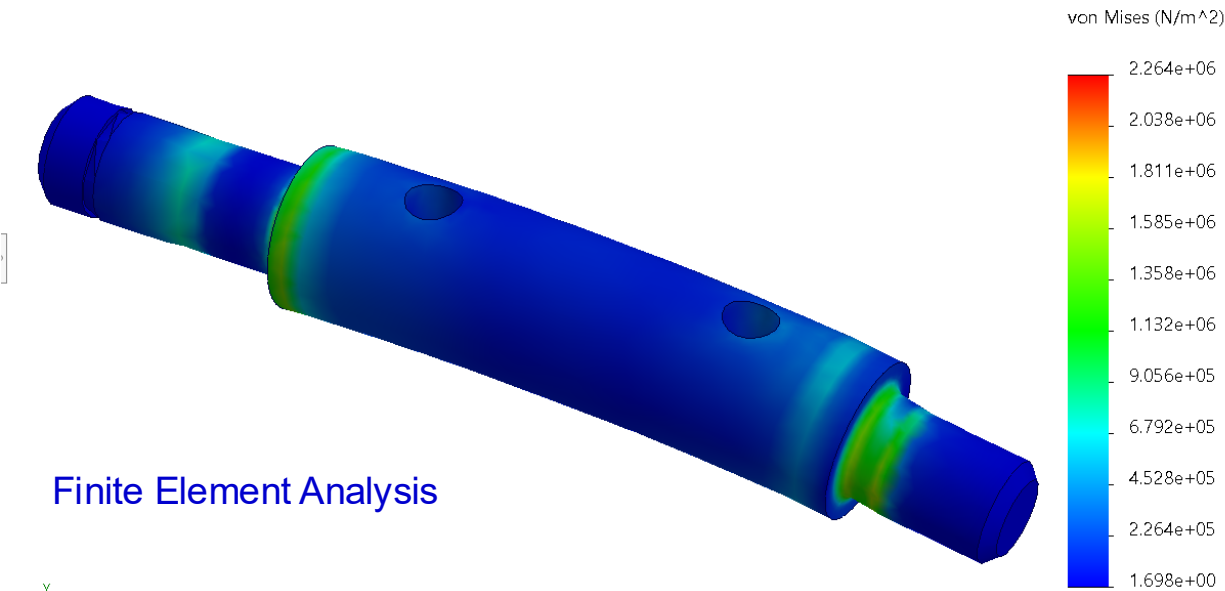
Would last about 3–4 tennis games

Critical Components: Wheel Shafts and Chassis Base Plate

FEA Analysis on Driving Wheel Shaft

Model name: Driven Shaft Assembly
Study name: Static 1(-Default-)
Plot type: Static nodal stress Stress1
Deformation scale: 30,655.3

Yield Stress: 620 MPa



Finite Element Analysis

von Mises Stress (MPa)	Principle Stress (MPa)	Shear Stress (MPa)
2.264	1.808	1.25

FEA	Hand calculated
Yielding (vm)	Shear
273	78
	Bending
	28

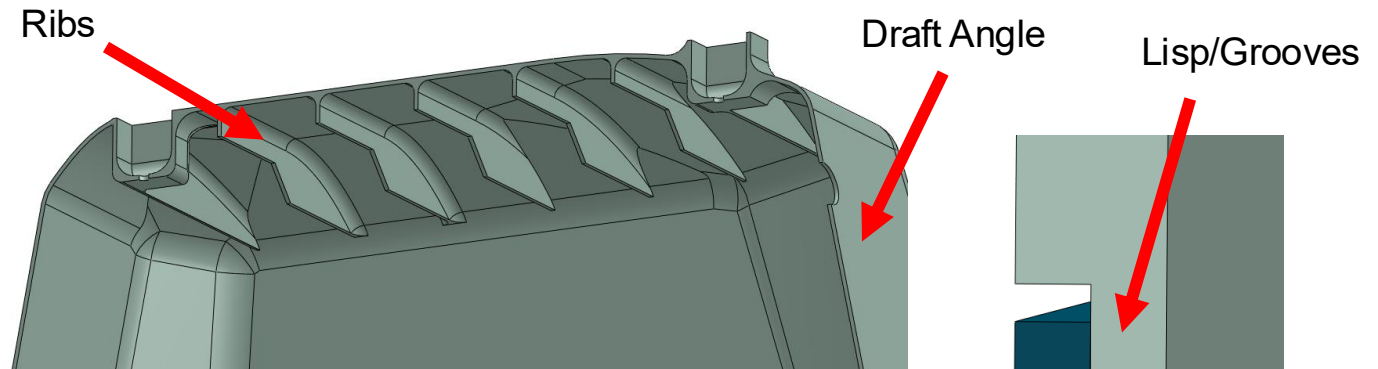
Fatigue
No fatigue failures due to fatigue limit around 500 MPa

Creep
No creep failures due to operational temperature under 400°C

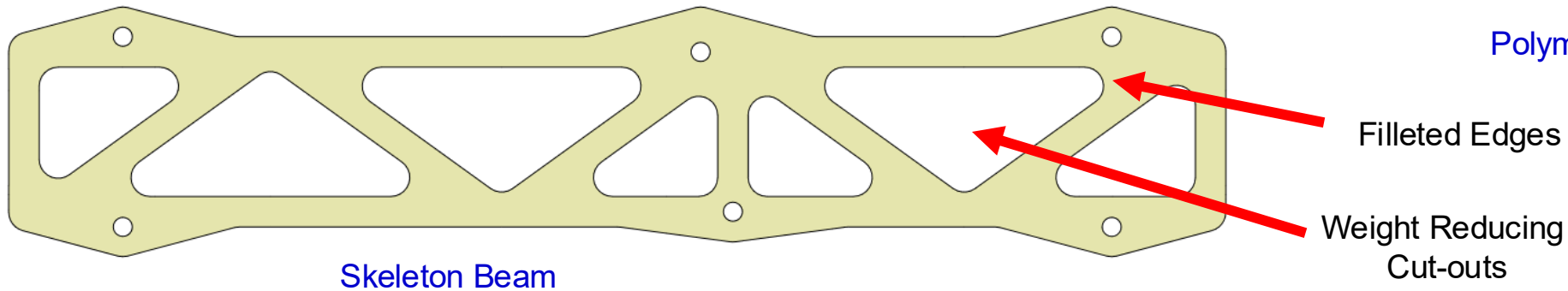
Key Components	Materials	Manufacturing Method
Shaft	AISI 4140 [low-alloy steel]	CNC turning
Outer casing	ASA [thermoplastic]	Injection molding
Structural frame [chassis, struts]	A356-T6 [aluminium]	CNC milling

Key Considerations:

- Aluminium frame components are duplicated and skeletonised
- Polymer shells are drafted by a minimum of 2°
- Polymer shells have ribs along its edges
- Designed with lips/grooves and no undercuts



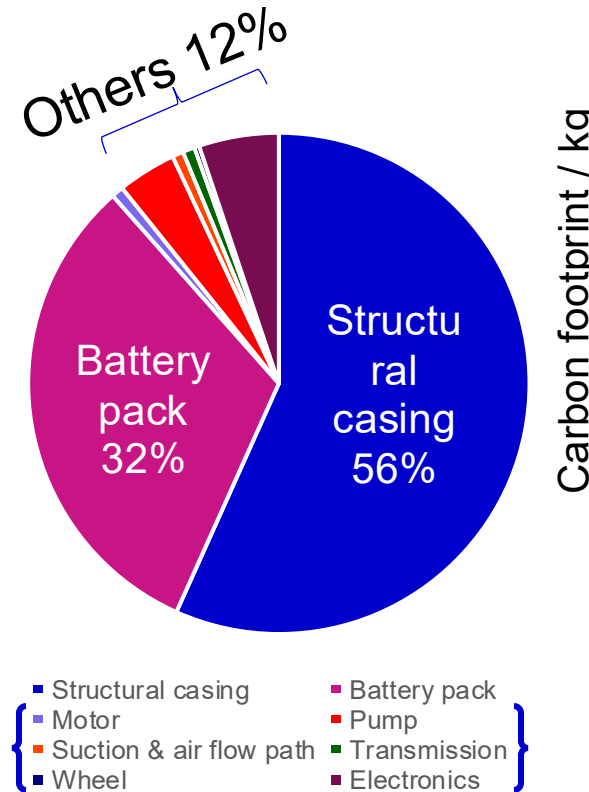
Polymer Casing



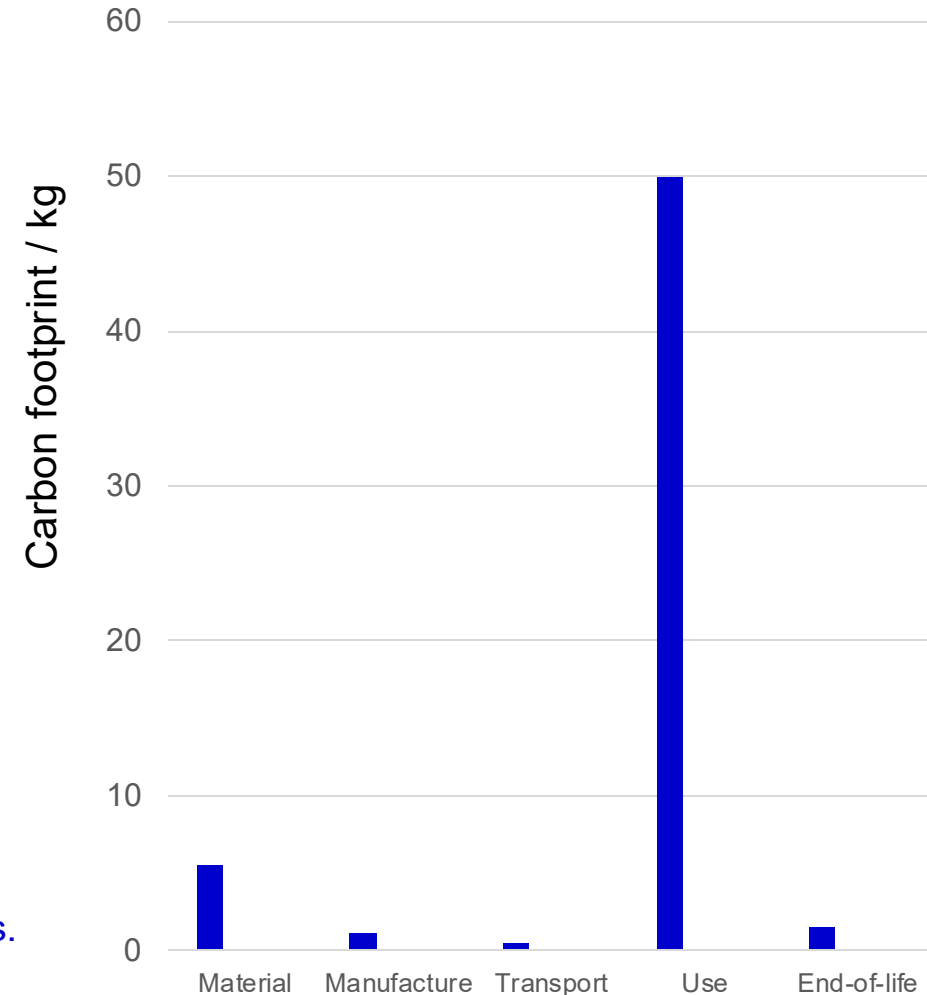
Skeleton Beam

1. Aluminum dominates manufacturing CO₂ because it's ~5/6 of total mass.
2. Impact reduction: switch non-structural parts to recyclable thermoplastics; keep plastic/metal easy to separate for end-of-life recycling.
3. Battery is a major contributor due to required range; future option: solar-assisted charging → smaller pack.
4. Use phase dominates total footprint → priority is cutting operating energy (e.g., reduce pressure losses → smaller, lower-power pump).

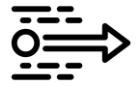
$$CO2_{use} = (P_{avg} * t) * Grid Factor .$$



Carbon footprint (kg) by parts.



Carbon Footprint (kg) in product life cycle (10 years).



Braking & acceleration

Fine adjustments by Control Team



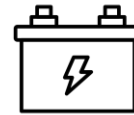
Corner balls

Leave for manual pickup during set breaks



Durability

$K_{IC} = 4 \text{ MPa m}^{1/2}$; Foam cover at the dock



Battery cooling

Base plate for air cooling; sufficient fan space



Tennis court types

Hard ✓; Grass & Clay ○



Fail-safe

Latch remains extended; storage sensor

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Summary

Our SHOBOT has accomplished...

Sustainable

Total CO₂-eq 58 kg per 10 years,
much below our target

Modular

Layered design, optimised for
ergonomics for maintenance

Non-disruptive

10s pick up time, no loud noise,
aesthetically pleasing

Reliable

Shaft bending safety factor 28.07;
Belt power safety factor 9.8

